# Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning

Zhenchao Tang*[1], Fang Wang*†[1], Haohuai He[1], Jiale Zhou[1], Tianxu Lv[1], Jun Zhu[1], Shouzhi Chen[1], Minghao Yang[1], Yu Wang[1], Jiayang Wu[1], Yidong Song[2], Yaokun Li[3], Jiehui Huang[4], Jun Zhou[5], Bing He[1], Jianhua Yao†[1]

[1]Tencent AI for Life Sciences Lab [2]Nankai University [3]Sun Yat-sen University
[4]The Hong Kong University of Science and Technology [5]Tsinghua University
*Equal contribution †Correspondence: avonwanghit@gmail.com, jianhua.yao@gmail.com

NEURAL INFORMATION PROCESSING SYSTEMS

Tencent AI for Life Sciences Lab

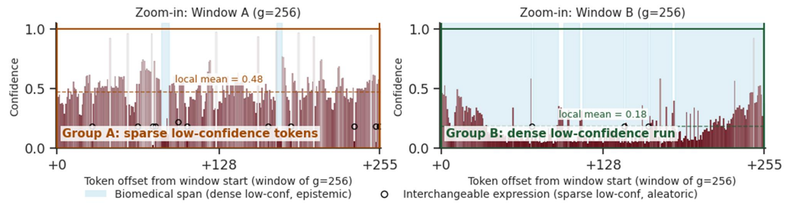
NeurIPS 2026 arxiv.org/abs/2511.21075
Project and code: shapsider.github.io/BFT

## 1 Problem & challenges

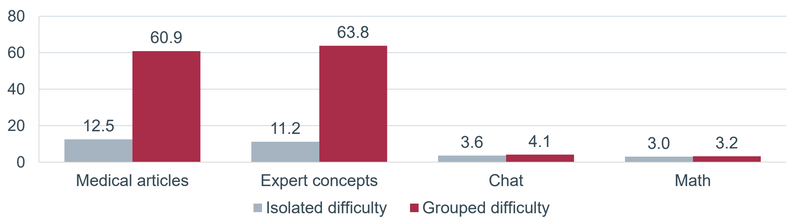
Biomedical AI must learn unfamiliar gene names and how biological processes work. Confidence-weighted fine-tuning (DFT) gives less emphasis to hard-to-predict words and can overlook difficult passages. Later training receives limited feedback, making this first learning stage important.

### Observation: difficult words occur in groups

Model predictions on medical articles, using nearby text



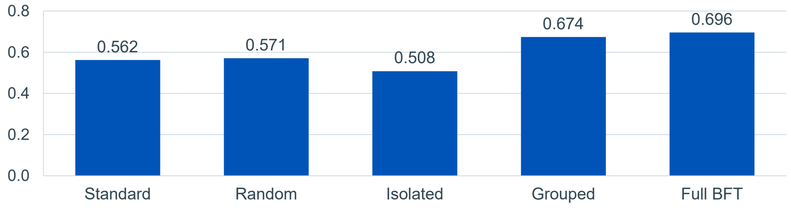

Share of biomedical terms / expert-marked concepts (%)




Groups of difficult words contain many biomedical concepts. Expert annotations support this pattern, while chat and math texts show little change in biomedical-term share. This suggests knowledge gaps, although prediction difficulty does not directly measure what the model knows.

### Which difficult content should training retain?

Cell-response prediction accuracy, same training settings




Keeping grouped difficult content gives 0.674 accuracy, compared with 0.571 for keeping the same amount of randomly chosen content. BFT uses gradual emphasis rather than a fixed keep-or-discard rule.

## 2 Motivation

A difficult word should be interpreted in context. A whole difficult passage may contain important biomedical knowledge that the model needs to learn. BFT therefore makes two connected decisions: which words should influence learning, and which training examples deserve more attention.

## 3 How Balanced Fine-Tuning works

BFT compares how hard each piece of text is to predict with the surrounding passage. It then gives more attention to training examples containing difficult passages.

**Look within an answer**

Read a word together with nearby text.
Keep more attention on difficult biomedical passages.

**Look across examples**

Find the hardest passage in each example.
Give harder examples more attention, within a limit.

**Learn difficult biomedical knowledge while limiting the influence of isolated outliers.**

### 1. Compare a word with its surroundings

DFT gives less emphasis to every hard-to-predict word. BFT first checks the surrounding passage. A word inside an equally difficult biomedical passage retains more learning signal than it would under DFT.

| Context | DFT | BFT |
|---|---|---|
| One difficult word, easy surroundings | Reduces its contribution | Keeps its emphasis low |
| **Many difficult words, difficult surroundings** | **Reduces their contribution** | **Restores more learning signal** |

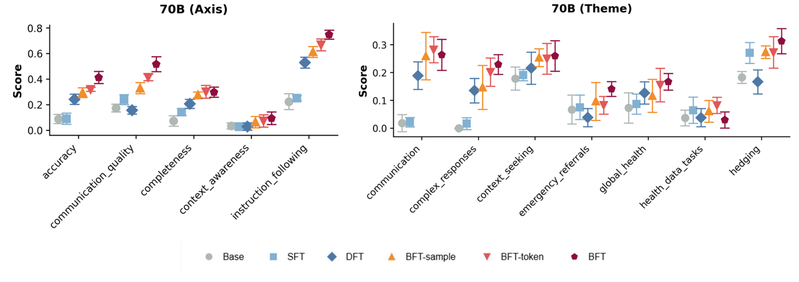
### 2. Give harder examples more attention

BFT finds the hardest local passage in each training example. Examples containing harder passages receive more emphasis. This extra emphasis is limited so that a difficult example cannot receive an unbounded weight.

### 3. Combine both decisions while learning

The first decision controls which words influence learning inside a response. The second controls how much the whole response contributes. Together, they focus training on difficult biomedical content while limiting isolated outliers.
We test both decisions under matched settings: 48,503 training examples and two passes through the same data.

### HealthBench Hard (70B) & GeneAgent



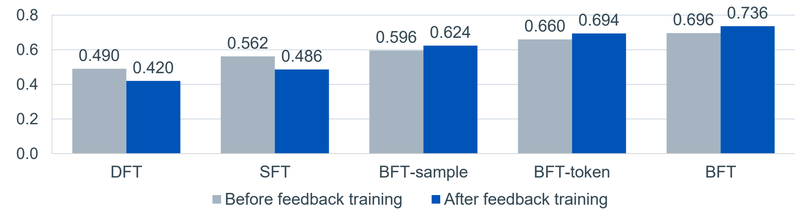

BFT improves medical scores across three model sizes and has the best overall GeneAgent text-overlap score among the aligned open-weight models. Error bars show variation across repeated answers.

## 4 Experimental results

We compare standard fine-tuning (SFT), DFT and BFT under matched settings, then add correct-answer feedback training (GRPO).

### Predicting how cells respond to chemicals

Predict gene activity: up, down or unchanged
Feedback training on 45 cell lines, testing on 5

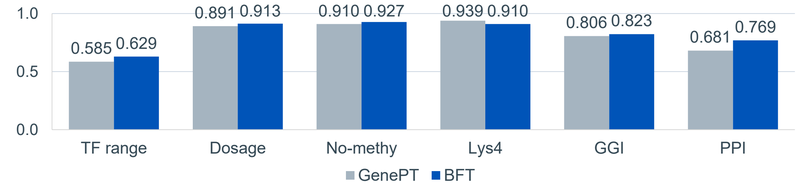



**+13.4 percentage points over SFT**
**+4.0 points after feedback training**

The two BFT parts help separately. SFT and DFT decline after feedback training. BFT's advantage over Gemini-2.5-Flash is not statistically clear.

### Turning gene descriptions into useful numbers

BFT writes gene descriptions. A text encoder converts them into numbers used to predict gene properties and group cells.

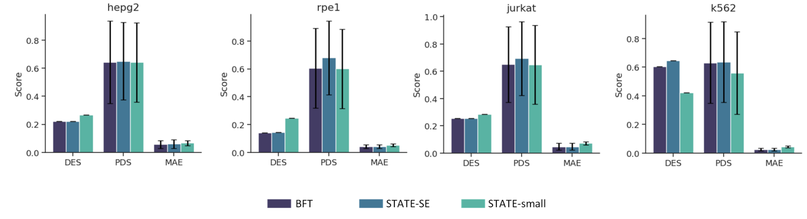



Average gene prediction score: BFT 0.829 vs GenePT 0.802. GenePT remains higher on Lys4. Both use the same text encoder, while the descriptions come from different sources.

**Grouping cells by type (higher scores are better)**

| Method | ARI | AMI | ASW |
|---|---|---|---|
| scGPT | 0.26 | 0.32 | 0.039 |
| GenePT | 0.30 | 0.46 | 0.041 |
| SFT | 0.12 | 0.25 | 0.021 |
| **BFT** | **0.34** | **0.58** | **0.061** |

### Predicting responses in single cells




Comparable to standard STATE inputs on four datasets. BFT writes profiles without training on these datasets, while the downstream STATE prediction model is still trained.